



Indian Institute of Technology
Kharagpur

S&P Global

CRISIL

TREMOR

Text-driven Risk Engine for Market Observation & Response

Converting unstructured financial news into explainable risk signals, market scenarios, and credit risk insights for faster, data-driven decisions.



Real-time
Event Understanding



Actionable
Risk Signals



Market & Credit
Impact Analysis



Explainable
& Transparent





The Problem & Our Approach

Financial markets react to real-world events. The challenge is to convert the massive, unstructured news flow into timely, explainable and actionable risk insights.



The Problem

Global events, policy decisions and supply chain shocks move markets, but the information is unstructured, noisy and overwhelming.

A constant firehose of unstructured news

Reuters Feb 23, 2022
US announces new Russia sanctions
Washington unveils fresh measures targeting Russian banks and energy sector...

Financial Times Mar 4, 2022
European banks increase risk exposure
Lenders face higher counterparty and market risks amid escalating conflict...

WSJ Mar 7, 2022
Supply chain disruptions intensify
Global supply chains face renewed pressure as trade routes are disrupted...

⋮

Key challenges for risk teams



High volume & noise
Tens of thousands of news articles daily across sources, languages and regions.



Slow and manual analysis
Analysts spend significant time reading, tagging and interpreting events.



Hard to quantify impact
Linking events to market and credit impact requires domain expertise and structured analysis.



Lack of transparency
Black-box signals are hard to trust and act on in regulated environments.



Our Approach – TREMOR

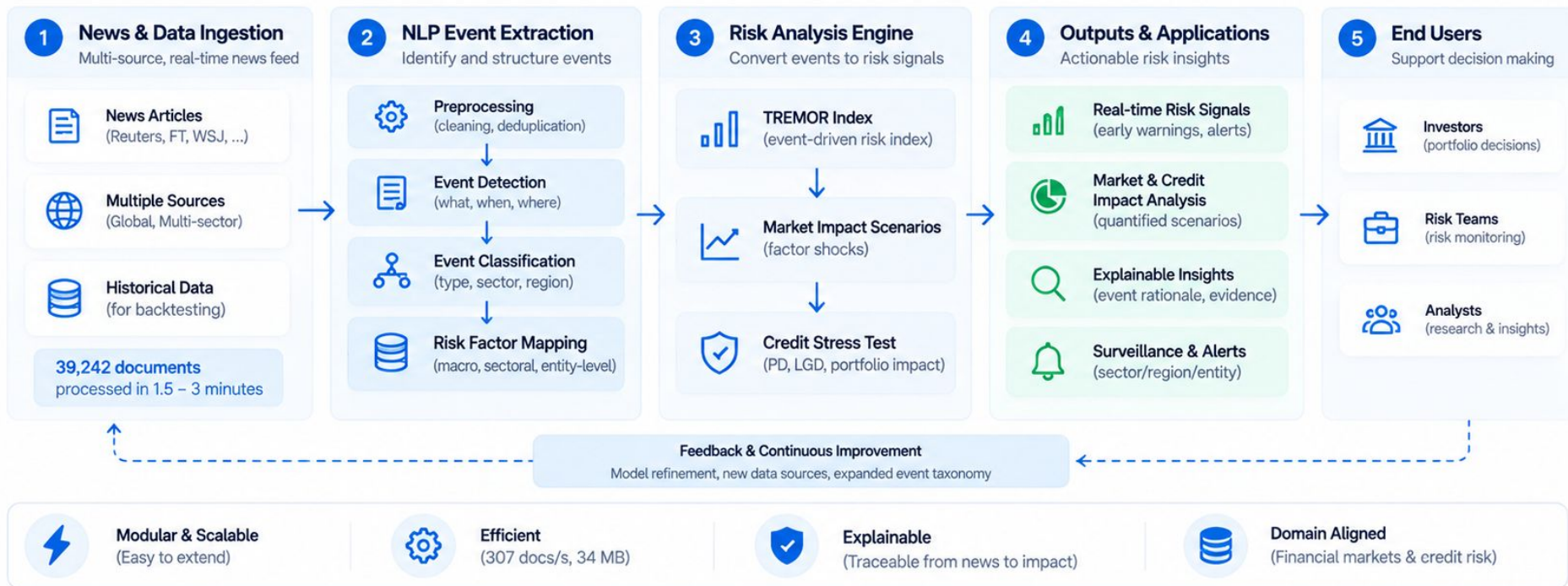
Convert unstructured news into explainable risk signals, market scenarios, and credit risk insights.





System Design – From News to Risk Insights

— A modular, end-to-end pipeline that converts unstructured news into structured events, risk signals, market scenarios and credit risk insights.





Implementation Highlights – Core Components of TREMOR

A modular pipeline that converts unstructured news into risk signals, market scenarios and credit risk insights.

1 NLP Engine – From News to Structured Events

Detects and classifies financial events with context using a fine-tuned language model.



News Articles
(Reuters, FT, WSJ, ...)



Preprocessing
Cleaning, deduplication,
segmentation



Event Detection & Classification
(what, when, where, type)



Entity & Risk Factor Mapping
(sector, region, asset class)



Structured Events
(for downstream modules)

Event Type Classification (Macro F1)

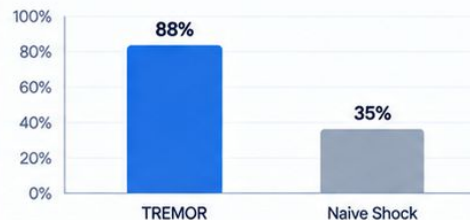


- 39,242 documents processed
- 307 docs/second
- 1.5 – 3 minutes total processing time
- ~4.2 ms per headline (CPU)
- 34 MB model size

2 TREMOR Index – Event-driven Risk Signals

Converts structured events into a time-series index capturing market risk factor movements.

TREMOR Index vs Naive Shock – Risk Factor Direction Accuracy



Captures key market risk factors

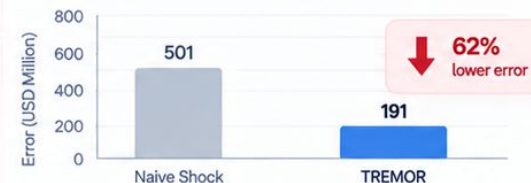
- Equity Markets
- Interest Rates
- Commodities
- Credit Spreads
- FX
- Volatility

3 Credit Stress Test – From Events to Portfolio Impact

Translates market scenarios into potential credit losses and capital impact for financial institutions.



Mean P&L Error vs Naive Baseline



Portfolio & Capital Impact (Ukraine Escalation Example)

Impact Score
7.7 → 9.9
(escalation)

CET1 Ratio
13.5% → 9.68%
(at peak)

Risk Factor Directions
8 / 9
correct



Technology Stack



Python
(Core pipeline)



Hugging Face
(NLP models)



PostgreSQL
(Data storage)



FastAPI
(API & services)



Streamlit
(Dashboard)



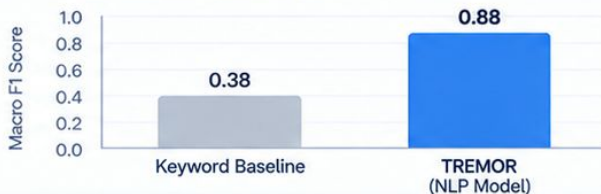
Key Results – Strong Performance Across All Modules

TREMOR outperforms naive baselines across event classification, risk inference, market impact and credit risk analysis.

1

Event Classification Performance

TREMOR significantly outperforms keyword-based baseline in identifying event types from financial news.



2.3×

higher Macro F1
than keyword
baseline

2

Risk Factor Direction Accuracy

TREMOR captures the correct direction of market risk factors in 88% of cases, vs 35% for naive shock.



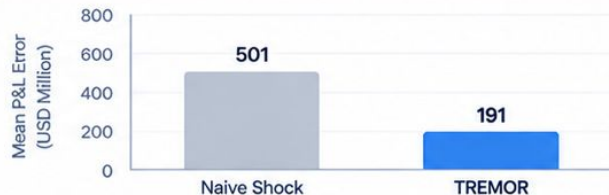
2.5×

higher direction
accuracy

3

Market Impact – Lower P&L Error

TREMOR produces significantly more accurate market impact estimates with 62% lower mean error than the naive baseline.



62%
lower mean
P&L error

Better than naive in
20 out of 21
crises

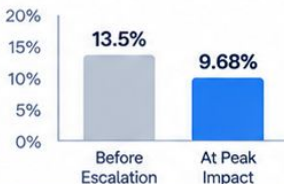
4

Credit Risk – Ukraine Escalation Case Study

In the Ukraine escalation scenario, TREMOR captures rising impact and credit stress for financial institutions.



CET1 Ratio (At Peak Impact)



8 / 9

risk factor
directions
correct



Key Takeaways



88%
risk factor directions correct
vs 35% naive



62%
lower mean P&L error
(\$191M vs \$501M)



0.88
macro F1 for event classification
vs 0.38 baseline



20 / 21
crises better than naive
baseline



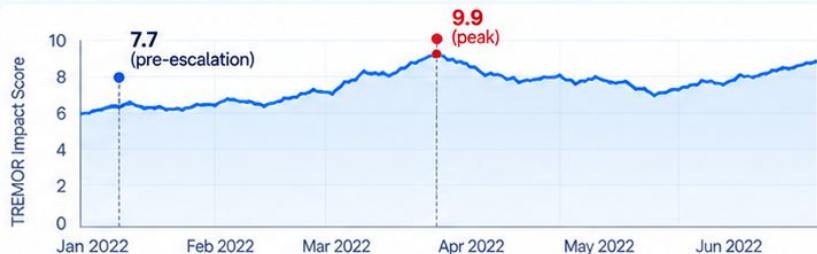
From News to Decisions – Ukraine Escalation Case Study

TREMOR converts real-world events into quantified risk signals, market impact estimates and credit portfolio insights.

1

Event Timeline and TREMOR Impact Score

TREMOR captured the escalation in real time, with impact score rising from 7.7 to 9.9.



2

Simulated Credit Impact (Representative Portfolio)

Using TREMOR scenarios, we estimate potential portfolio losses and capital impact.

Portfolio P&L (USD Million)



Simulated CET1 Ratio (Representative Bank)



3

Risk Factor Movements (Direction Accuracy: 8/9)

TREMOR correctly identified the direction of 8 out of 9 key risk factors.

Risk Factor	Actual Movement (Mar-Apr 2022)	TREMOR Predicted Direction	Correct?
Oil Prices	↑	↑	✓
Natural Gas Prices	↑	↑	✓
Equity Markets	↓	↓	✓
Sovereign Spreads	↑	↑	✓
USD Index	↑	↑	✓
Interest Rates	↑	↑	✓
Credit Spreads	↑	↑	✓
Volatility (VIX)	↑	↑	✓
Gold Prices	↑	↓	✗

4

Example Events and Model Interpretation

TREMOR provides interpretable event classification and risk factor links.

Date	Headline (shortened)	Event Type	Key Risk Factors	TREMOR Impact
24 Feb 2022	Russia launches military operation in Ukraine	Military Conflict	Oil ↑, Gas ↑, Equity ↓	8.9
02 Mar 2022	Western nations announce new sanctions on Russia	Economic Sanctions	FX ↑, Credit Spreads ↑, Equity ↓	9.4
08 Mar 2022	European gas prices surge to record high	Commodity Shock	Gas ↑, Inflation ↑, Interest Rates ↑	9.6
15 Mar 2022	Talks between Russia and Ukraine show limited progress	Geopolitical Tension	Volatility ↑, Equity ↓	9.2



Key Takeaway

TREMOR translates news events into explainable risk signals, enabling early detection of market stress, quantification of portfolio losses and assessment of capital impact for financial institutions.



Key Takeaways and the Way Forward

TREMOR demonstrates the potential of text-driven risk intelligence for real-world market and credit risk monitoring.

1

Key Takeaways

TREMOR effectively converts news and textual data into quantifiable risk signals.



Captures Real-World Impact

Effectively identifies and quantifies market and credit risk from news events.



Interpretable Risk Signals

Provides direction, key risk factors and explainable insights for better decisions.



Modular and Scalable

Designed to integrate multiple data sources and scale to real-world use.



Strong Domain Relevance

Applicable for financial institutions, credit rating and investment analysis.

3

Potential Impact

TREMOR can create value for multiple stakeholders in the financial ecosystem.



Financial Institutions

Early risk detection and better portfolio management.



Credit Rating Agencies

Enhanced credit assessment with real-time insights.



Investment Firms

Data-driven decisions and improved risk-adjusted returns.



Policy Makers

Monitor systemic risks and emerging global events.

2

Future Scope

We plan to further enhance TREMOR to make it more robust and production-ready.



Broader Asset Coverage

Include more asset classes (equities, bonds, commodities, FX).



Advanced Modeling

Incorporate nonlinear models, sector linkages and feedback effects.



Real-Time Deployment

Build a production-ready pipeline with streaming ingestion and alerts.



Wider Data Sources

Integrate additional news providers, company filings and structured data.

4

Project Links and Team

Explore the code and live demo. Team DASK – IIT Kharagpur.

Project Links



GitHub Repository



Live Demo



Team DASK (Team ID: 178814)

Shubham Suman (Leader)

Anurag Ranjan

Krupal Thakre

Durgesh Kumar